



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC AUTONOMOUS)

B.Tech III Semester Supplementary Examinations Nov/Dec-2025

Course Name: Analog & Digital Electronics

(Common for CSE & IT)

Date: 26.11.2025 FN

Time: 3 hours

Max.Marks: 70

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions

Each question carries TWO marks.

10x2=20M

1. Define rectifier. 2 M
2. Draw the characteristics of Zener Diode. 2 M
3. What is the need for biasing? 2 M
4. How the UJT acts as a Relaxation Oscillator? 2 M
5. Draw the characteristics of N-channel Depletion type MOSFET. 2 M
6. Give the properties of Binary codes. 2 M
7. Define Multiplexer. 2 M
8. Write the truth table for EX-OR gate. 2 M
9. Write the characteristic Equation of JK Flip-flop. 2 M
10. Define Latch. 2 M

PART-B

Answer the following. Each question carries TEN Marks.

5x10=50M

- 11.A). Draw the band structure of open circuit PN-Junction and the expressions for contact difference of potentials. 10M

OR

11. B). With the help of Circuit and wave form explain the working of Half wave and full wave rectifiers and derive the expressions for form factor and ripple factor and efficiency? 10M

12. A). Demonstrate the input and output configurations of BJT in common base configuration. 10M

OR

12. B). Explain in detail about the working principle and operation of UJT. 10M

13. A). Describe the construction and working of Enhancement type and depletion type MOSFET. 10M

OR

13. B). List the properties of Boolean Function and explain. 10M

14. A). Implement the following using NAND NAND logic 10M
(i) $AB+D(C+AB)+CD$ (ii) $(X+Y)+(Y+Z)$

OR

14. B). With a neat sketches explain the operation of Comparator circuits. 10M

15. A). Design a counter with the following repeated binary sequence 0,1,2,3,4,5,6 using JK Flipflop. 10M

OR

15. B). Convert a SR Flip-flop to a T-Flip-flop. 10M
